

Towards Semi-automatic Agentic Performance Engineering

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GitHub Next

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GitHub Next investigates the future of software development.

We are a team of researchers and engineers at GitHub, exploring things beyond the adjacent possible. We prototype tools and technologies that will change our craft. We identify new approaches to building healthy, productive software engineering teams.

We live in incredible, disturbing times

- **AI, indexing and traditional dev tools** make us all magicians when it comes to code



We live in incredible, disturbing times

- **AI, indexing and traditional dev tools** make us all magicians when it comes to code
- But the magic power comes with many difficulties!



AI Dev Tools are changing fast

- **2022: GitHub Copilot Completions** launched.
 - 1.8M+ paying users, 50%+ of users' code is written by Copilot
- **2023: More AI developer features launched**
 - Summarization, advice, code review, IDEs and others
- **2024: GitHub Copilot Workspace, Cursor Composer, Devin** launched
 - Shift to task-orientation, Cursor Composer brought task-orientation to the IDE
- **2025: Cursor/VS Code Agent-Mode, Claude Code and others** launched
 - Increases the available automation
- **2025: Automated app dev** starts to take off
 - GitHub Spark, Bolt, Gemini, Lovable, Cogna, v0, ...
- **2026: Continuous AI. Event-triggered, proactive agentic. Semi-automatic software dev**

Continuous AI
implemented by
Agentic Workflows
enables
Continuous Engineering

Continuous AI

Inner Loop, Outer Loop

- Continuous AI is any automated use of AI to facilitate software engineering, on any platform, anywhere
- <https://githubnext.com/projects/continuous-ai/>

CI/CD → CAI

- CI/CD revolutionized software engineering
- AI in the outer-loop?
 - Continuous Security
 - Continuous Documentation
 - Continuous Triage
 - Continuous Facilitation + Communication
 - Continuous Code Improvement
 - Continuous Agent Improvement (Monitoring, Improvement, Learning)
- Gives rise to an “AgentOps” mindset

Characteristics of Continuous AI

- Proactive
- Automatable
- Repetitive
- Collaborative
- Integrated
- Auditable
- Organization-specific
- Event-triggered

AI for collaboration, not just individual productivity

- Automate tasks that benefit *teams*
- Aim: more efficient, enjoyable collaborative environment
- Aim: humans and AI, working happily together
- Aim: mitigate team-negative AI with team-positive AI

Ingredients of Continuous AI

1. Event-based triggering
2. Execution environment + Inference
3. An LLM programming framework, e.g., [GenAIScript](#), [llm](#), [LangChain](#), etc.

OR: A hosted agentic framework

OR: An agent platform

OR: A “virtual contributor” platform

Continuous AI
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GitHub Agentic Workflows

What is an agentic workflow?

- [GitHub Agentic Workflows](#)
- An **agentic coding agent** = **LLM + tools + plan/todos + loop + task**
 - Copilot CLI / Claude Code / OpenAI Codex/...
 - Tools pluggable using [Model Context Protocol](#).
 - Enable **natural language programming**
- An **agentic workflow** is a repeatable task for a coding agent
 - “Frontmatter + markdown”
 - Frontmatter = Triggers, Tools, Engine, Security, Safe Outputs

Example - Weekly Researcher

```
---
on:
  schedule:
    - cron: "0 9 * * 1"    # Every week, 9AM UTC, Monday
  workflow_dispatch:

safe-outputs:
  create-issue:

tools:
  web-search:

---

# Agentic Weekly Researcher (Ruby)

Do a Assumes GitHub in ${github.repository} repository, and the related industry in general.

- Read self-reports of the latest code, issues and PRs for this repo.
- Read latest trends and news from the software industry news source on the Web.

Create a new GitHub issue containing a markdown report with interesting information related to this software project, related products and competitive analysis, related opportunities, business analysis, enjoyable anecdotes.

Self-reporting

> NOTE: Include a link like this at the end of the report:

...

> AI-generated content by ${github.workflow}}, may contain mistakes.
...
```

Triggers

Safe output specs

Tools

Actions-familiar

NL programming =
deliberate ambiguity =
genuinely useful
ambiguity

Assumes GitHub

Self-reporting

Example - Issue Triage

```
---
on:
  issues:
    types: [opened, reopened]
```

```
safe-outputs:
  add-comment:
```

```
---
```

```
# Agentic Issue Triage
```

Your task is to analyze issue `#{ github.event.issue.number }` and perform some initial triage tasks related to that issue.

Analyze the issue content. Write notes, ideas, nudges, resource links, debugging strategies and/or reproduction steps for the team to consider relevant to the issue. Select appropriate labels already used in the repository and apply them to the issue.

Finalize an output report with your analysis and add it as an issue comment to the issue.

> NOTE: Include a link like this at the end of your issue comment:

```
```markdown
```

> AI-generated content by `#{ github.workflow }`, may contain mistakes.

```
```
```


Agentic Workflows - The ultimate vision

1. You add `.github/workflows/agent.md`
2. The agentic workflow starts working

That's all

Agentic Workflows - The mid-step vision

1. You add `.github/workflows/agent.md`
2. It is transformed to `.github/workflows/agent.lock.yml`
3. The agentic workflow starts working

That's all

Agentic Workflows - Getting Started

Samples

```
gh extension install githubnext/gh-aw
```

```
gh aw add githubnext/agentics/daily-test-improver
```

```
<secret+commit+push>
```

```
gh aw run daily-test-improver --repeat 30
```

Agentic Workflows - Design Principles

1. The markdown is **actions-familiar**
2. The defaults are **simple**
3. The emphasis is on **natural language markdown**
4. The processing assumes **agentic processor + MCPs**
5. The markdown can include **shared components**
6. Safety, Safety, Safety
7. The end-to-end is **secure but powerful**

Agentic Workflows - Safety, Safety, Safety

1. Execution is VM-sandboxed (GitHub Actions VMs)
2. Overall permissions are minimized
3. Agentic steps run essentially read-only
4. All outputs are “in the intended channel” (“safe outputs”)
 - a. blast radius greatly reduced
 - b. checked by AI for safety
 - c. human oversight guaranteed
5. Outgoing network is firewalled and controlled

Agentic Workflows - GitHub Actions - The Place to Be!

1. GitHub Actions is the place that “knows” how to restore, build and test your code
2. Cheap compute, packages, caching, containers, security, monitoring, cost-control, networking, integrated auth, secrets, environments, ...
3. Agentics + GitHub Actions is one place to explore agentic engineering, today



Agentic Workflows - Combining Hard + Soft

Agentic Workflows can include

1. deterministic steps (trad GitHub Actions YML)
2. agentic steps (coding agents)
3. deterministic steps (custom safe outputs)

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Continuous Test Engineering

Continuous Test Improvement

Sample workflow

```
gh extension install githubnext/gh-aw
```

```
gh aw add githubnext/agentics/daily-test-improver
```

```
<secret+commit+push>
```

```
gh aw run daily-test-improver --repeat 30
```

OR YOLO YOLO YOLO YOLO

```
gh aw trial githubnext/agentics/daily-test-improver --clone-repo  
Comcast/gaad --host-repo githubnext/gaad --auto-merge-prs --repeat 30
```

Continuous Test Improvement

Samples:

- [FSharp.Control.AsyncSeq](#) 10/11 PRs merged
- [FsMath](#) 38/44 PRs merged, ~99% coverage
- [Go Audio Decoder](#),
- [Oxpecker Web Framework](#), 50% → 87% coverage in a few hours
- ...

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Continuous Perf Engineering

Continuous Perf Engineering: A Grand Challenge

- Perf engineering is everywhere, but hugely neglected
 - It is **hard, heterogeneous, multi-skilled, nuanced, requires careful judgement**
 - It involves **Research, Planning, Measuring, Coding, Tuning, Re-measuring, Testing**
 - Must stay aligned with human/business goals
 - Takes investment, thought, care

Continuous Agentic Perf Engineering

- Agentic platforms open the door to a whole new world of semi-automated software engineering
- Agentic workflows can
 - Do depth research
 - Do depth planning
 - Align with human-declared priorities
 - Prepare VMs for depth perf engineering
 - Equip LLMs with tools for coverage, profiling, telemetry and log analysis
 - Self-observe and self-improve
 - Invest in performance engineering efficiency

Continuous Perf Improvement - Example

Daily Perf Improver

```
gh extension install githubnext/gh-aw
```

```
gh aw add githubnext/agentics/daily-perf-improver
```

```
<secret+commit+push>
```

```
gh aw run daily-perf-improver --repeat 30
```

OR YOLO YOLO YOLO YOLO

```
gh aw trial githubnext/agentics/daily-perf-improver \  
  --clone-repo Comcast/gaad \  
  --host-repo githubnext/gaad \  
  --auto-merge-prs --repeat 30
```

Continuous Perf Engineering: The Approach

- Phase 1: Research + Plan
- Phase 2: Build Step Inference + 1-5 Guides
- Phase 3 Repeated: Choose/Dev/Measure/Report

NOTE: many other variations possible

Continuous Perf Engineering: The Possible

- [FSharp.Control.AsyncSeq](#)
 - ~2K LOC async sequence library
 - Long standing perf bugs found and addressed
 - 7/7 PRs accepted
- [FsMath](#)
 - ~3K LOC math library
 - 8/12 PRs accepted, 4 outstanding
 - E.g. “[Optimize vector x matrix multiplication with SIMD](#)”
- [Hono](#)
 - ~15K LOC high-performance web framework
 - ~20 PRs prepared
 - [Invitation to maintainers to assess](#)

Continuous Perf Engineering: The Potential

- Every top Python, Rust, .NET, Java library
- C++ math libraries (e.g. LibRapid)
 - (but automated measurement is hard)
- GitHub Internal Services
 - e.g. hosted compute watchdog and other cloud services
 - transact vast amounts of data
 - high operational costs

Continuous Perf Engineering: The Delusional/The Challenge

- [Z3Prover/z3](#)
 - 16/18 PRs rejected - evidence not properly collected, results hallucinated/fabricated
 - “Looked in right places” - e.g. hashing performance
 - Was controversial with some maintainers
- Linux Kernel
 - Requires a whole foundation to engineer automated testing
 - Hardware and operational needs very difficult
- [github.com](#)
 - Multiple very large codebases
 - Multi-layered, multi-service backend
 - A grand challenge to automate the analysis and improvement of such a service

Continuous Perf Engineering: Aligning

- “In production most the problems we have are memory”
- A little insight text goes a long, long way
- Requirements gathering & elaboration is subtle, continuous, and always conversational

Continuous Perf Engineering: Continuous Improvement

- Latency on a massive-scale web site/service
- Every learning is a constraint to guide future success
- Example: Learning disjunctive guides
- A form of agentic memory

Continuous Perf Engineering: Agents need data too

- So far not focused on telemetry-guided performance engineering
- Agents “give up” and ask for production data
- MCPs for telemetry services are now commonplace

Continuous Perf Engineering: Linear, not Swarms

- Swarms are cooperating, concurrent agents with different roles
- They create many conflicts, duplication, overlap and need careful coordination and planning
- Linear phasing is often more productive
 - Research → Plan → Prepare → Dev/Measure → Reproduce → Dev/Measure → Reproduce → Dev/Measure → Reproduce → Plan → Prepare →

Conclusion

The Path Ahead

- Every University should already have a 2nd year course on AI-assisted Software Development
- Every University should be planning a course on Semi-automated Software Development by 2026
- Expect huge disruption to software and team methodologies.
 - Agile begins to look wrong when agents have been coding all night
- Automated performance improvement will be A Grand Challenge, but is already real in some scenarios

Is Continuous AI the answer to AI slop?

- AI slop is real, can damage teams
- Today: much more human review
- Alternative: relentless, continuous AI to pre-judge, review, clean up, monitor, improve

Thank you